

APPENDIX S2: BEE SPECIES LIST WITH TRAITS

Table S2. List of bee species in the dataset, with trait assignments (includes females only). “Abd.” is the summed abundance over all sites of a given site type ($N = 4$ conventional farms, 5 organic farms, 7 natural sites). “Flight-season duration” is the value calculated from a sample of 30 independent phenology records, as described in Appendix S3; species for which <30 records were available are “NA” for this trait. “UK” = trait status unknown.

Species	Abd. on conventional farms	Abd. on organic farms	Abd. at natural sites	Inter- tegular distance (mm)	Median day of year of flight season	Flight- season duration	Nest location	Nesting behaviour	Sociality/ parasitic	Lecty
<i>Agapostemon texanus</i>		9	1	1.98	189	133.4	below	excavate	solitary	poly
<i>Andrena angustitarsata</i>	2	4		1.80	92	51.8	below	excavate	solitary	poly
<i>Andrena anisochlora</i>	1		1	1.93	72	NA	below	excavate	solitary	poly
<i>Andrena atypica</i>		1		1.38	98	NA	below	excavate	solitary	oligo
<i>Andrena auricoma</i>		1	3	1.62	99	45.3	below	excavate	solitary	poly
<i>Andrena candida</i>	1	11	1	1.65	127	86.6	below	excavate	solitary	poly
<i>Andrena cerasifolii</i>	1	2		2.65	113	82.8	below	excavate	solitary	poly
<i>Andrena cercocarp</i>	1	3		1.66	113.5	NA	below	excavate	solitary	poly
<i>Andrena chalybioides</i>		8		1.75	UK	NA	below	excavate	solitary	oligo
<i>Andrena chlorogaster</i>	9	23	6	1.24	91.5	50.2	below	excavate	solitary	poly
<i>Andrena cressonii</i>	5	3		2.15	111	NA	below	excavate	solitary	poly
<i>Andrena cuneilabris</i>		11		1.78	73	NA	below	excavate	solitary	oligo
<i>Andrena hypoleuca</i>			1	1.70	113	NA	below	excavate	solitary	UK
<i>Andrena misella</i>	1	1	1	1.93	88	NA	below	excavate	solitary	poly
<i>Andrena nigroaerulea</i>	1		4	1.91	92.5	NA	below	excavate	solitary	poly
<i>Andrena orthocarp</i>	10	19	1	1.48	81.5	NA	below	excavate	solitary	poly
<i>Andrena pallidifovea</i>		1		2.01	128	NA	below	excavate	solitary	oligo
<i>Andrena pensilis</i>		1		1.64	115	NA	below	excavate	solitary	oligo
<i>Andrena piperi</i>	16	55		1.38	92	66.5	below	excavate	solitary	poly

<i>Andrena semipunctata</i>		2		1.95	89	NA	below	excavate	solitary	oligo
<i>Andrena</i> sp. 1		3		1.44	NA	NA	below	excavate	solitary	UK
<i>Andrena suavis</i>			3	1.51	64.5	NA	below	excavate	solitary	oligo
<i>Andrena subaustralis</i>		2		2.16	98	47.9	below	excavate	solitary	oligo
<i>Andrena subchalybea</i>		11	4	1.56	116	NA	below	excavate	solitary	oligo
<i>Andrena subtilis</i>	1	7	2	2.20	UK	NA	below	excavate	solitary	poly
<i>Andrena w-scripta</i>		5	4	1.74	102	73.8	below	excavate	solitary	poly
<i>Anthidiellum ehrhorni</i>			1	2.39	150	NA	above	rent	solitary	poly
<i>Anthidiellum notatum</i>			2	2.15	230	111.0	above	rent	solitary	poly
<i>Anthidium mormonum</i>			1	2.29	UK	NA	above	rent	solitary	poly
<i>Anthidium utahense</i>			1	2.05	134.5	NA	above	rent	solitary	poly
<i>Anthophora californica</i>			6	3.58	123	NA	below	excavate	solitary	poly
<i>Anthophora curta</i>			2	2.68	214	116.9	below	excavate	solitary	poly
<i>Anthophora edwardsii</i>	2	5	3	4.08	90	NA	below	excavate	solitary	poly
<i>Anthophora pacifica</i>			1	4.24	60	NA	below	excavate	solitary	poly
<i>Anthophora urbana</i>	7	2	22	2.98	202	135.8	below	excavate	solitary	poly
<i>Ashmeadiella aridula</i>			2	1.25	217	85.8	UK	rent	solitary	poly
<i>Bombus californicus</i>	32	11	37	3.61	151	102.3	mix	rent	social	poly
<i>Bombus crotchii</i>		1		5.35	140.5	NA	below	rent	social	poly
<i>Bombus melanopygus</i>	13	40	57	3.86	104.5	75.5	mix	rent	social	poly
<i>Bombus vosnesenskii</i>	111	365	89	3.55	142	110.2	below	rent	social	poly
<i>Calliopsis fracta</i>			20	1.46	133	NA	below	excavate	solitary	oligo
<i>Ceratina acantha</i>	9	11	6	1.15	142	148.3	above	excavate	solitary	poly
<i>Ceratina arizonensis</i>		2	83	0.85	159	124.9	above	excavate	solitary	poly
<i>Ceratina dallatorreana</i>	1	4		1.29	174	98.5	above	excavate	solitary	poly
<i>Ceratina nanula</i>		7	12	1.02	167	NA	above	excavate	solitary	poly
<i>Ceratina punctigena</i>			1	1.62	131	NA	above	excavate	solitary	poly
<i>Ceratina sequoiae</i>			5	1.48	147	NA	above	excavate	solitary	oligo
<i>Ceratina</i> sp.	2	5	5	0.98	NA	NA	above	excavate	solitary	poly
<i>Ceratina tejonensis</i>			2	1.61	99	NA	above	excavate	solitary	poly

<i>Chelostoma phaceliae</i>			7	1.08	133	NA	above	rent	solitary	oligo
<i>Diadasia bituberculata</i>			1	3.30	127	NA	below	excavate	solitary	oligo
<i>Diadasia consociata</i>	1			1.85	199	96.6	below	excavate	solitary	oligo
<i>Diadasia enavata</i>		1		2.74	225	64.4	below	excavate	solitary	oligo
<i>Dianthidium ulkei</i>			1	1.89	217	NA	mix	rent	solitary	oligo
<i>Dufourea</i> sp. 2			1	1.61	NA	NA	below	excavate	solitary	oligo
<i>Eucera (Synhalonia) actuosa</i>	33	6	51	2.73	117	68.0	below	excavate	solitary	poly
<i>Eucera (Synhalonia) amsinckiae</i>	2	6	2	3.14	94.5	36.0	below	excavate	solitary	oligo
<i>Eucera (Synhalonia) frater</i>	4	7	14	3.90	126	50.4	below	excavate	solitary	poly
<i>Eucera (Synhalonia) lunata</i>		1	2	3.77	79.5	NA	below	excavate	solitary	poly
<i>Eucera (Synhalonia) stretchii</i>		1		3.07	118	NA	below	excavate	solitary	poly
<i>Eucera (Synhalonia) virgata</i>			2	3.02	UK	NA	below	excavate	solitary	poly
<i>Habropoda tristissima</i>		1	11	4.26	117.5	43.1	below	excavate	solitary	poly
<i>Halictus farinosus</i>	10	20	12	2.13	157	158.8	below	excavate	social	poly
<i>Halictus ligatus</i>	47	148	51	1.53	188	120.1	below	excavate	social	poly
<i>Halictus tripartitus</i>	283	350	333	1.20	169	116.5	below	excavate	social	poly
<i>Heriades occidentalis</i>			1	1.40	158	NA	above	rent	solitary	poly
<i>Hesperapis ilicifoliae</i>			3	1.43	150.5	NA	below	excavate	solitary	oligo
<i>Hoplitis albifrons</i>			2	2.63	133	NA	above	rent	solitary	poly
<i>Hoplitis fulgida</i>		1		1.51	100.5	NA	above	rent	solitary	poly
<i>Hoplitis grinnelli</i>			1	1.28	116	NA	above	rent	solitary	poly
<i>Hoplitis howardi</i>			2	1.15	151	NA	above	rent	solitary	oligo
<i>Hoplitis hypocrita</i>	2	2		2.59	130	NA	above	rent	solitary	poly
<i>Hoplitis producta</i>			12	1.27	126.5	NA	above	rent	solitary	poly
<i>Hoplitis sambuci</i>			1	2.23	137.5	NA	above	rent	solitary	poly
<i>Hylaeus calvus</i>			6	1.16	130	NA	above	rent	solitary	poly
<i>Hylaeus rudbeckiae</i>	3	35	12	0.97	136	NA	above	rent	solitary	poly
<i>Hylaeus verticalis</i>		1	3	1.44	120	NA	above	rent	solitary	poly
<i>Lasioglossum (Dialictus) incompletum</i>	234	185	139	0.94	171	100.1	below	excavate	social	poly

<i>Lasioglossum (Dialictus)</i> sp. B	14	7	14	1.30	189.5	99.8	below	excavate	social	poly
<i>Lasioglossum (Dialictus)</i> sp. C	8	13	7	1.23	113	NA	below	excavate	social	poly
<i>Lasioglossum (Dialictus)</i> sp. E	1	2	4	1.12	UK	NA	below	excavate	social	poly
<i>Lasioglossum (Dialictus)</i> sp. F		4	3	1.02	179.5	NA	below	excavate	social	poly
<i>Lasioglossum (Dialictus)</i> <i>tegulariforme</i>	16	31	42	0.88	172	96.3	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp.	4	58	3	1.30	NA	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. "red abd."			14	1.36	UK	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. A	1	3	7	1.48	143	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. B		2		1.44	66	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. C	3	3	5	1.64	97.5	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. D	5	3		1.68	UK	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. E	5	9	48	1.10	124	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. F	34	85		1.23	83	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. G			1	1.89	UK	NA	below	excavate	social	poly
<i>Lasioglossum (Evylaeus)</i> sp. I	5	3		1.65	82	NA	below	excavate	social	poly
<i>Lasioglossum (Lasioglossum)</i> <i>mellipes</i>		9	1	2.08	128	NA	below	excavate	solitary	poly
<i>Lasioglossum (Lasioglossum)</i> <i>titusi</i>	1	4	1	1.93	135.5	195.2	below	excavate	solitary	poly
<i>Megachile angelarum</i>		1	22	2.17	161	NA	above	rent	solitary	poly
<i>Megachile apicalis</i>	9	4	49	2.31	189	63.3	above	rent	solitary	oligo
<i>Megachile coquilletti</i>	1		10	3.36	218	102.6	above	rent	solitary	poly
<i>Megachile frugalis</i>			2	2.65	173	NA	above	rent	solitary	poly
<i>Megachile gentilis</i>		1	4	2.63	243.5	109.8	above	rent	solitary	poly
<i>Megachile montivaga</i>		1		2.68	189	NA	mix	rent	solitary	poly
<i>Megachile onobrychidis</i>		1		2.58	225	100.7	mix	rent	solitary	poly
<i>Megachile parallela</i>		1	1	3.27	218	NA	below	rent	solitary	poly
<i>Megachile rotundata</i>		1	1	2.20	176.5	170.2	mix	rent	solitary	oligo
<i>Melecta edwardsii</i>			1	3.56	98	NA	below	NA	parasite	NA

<i>Melissodes lupina</i>	6	1	13	2.20	216	85.9	below	excavate	solitary	oligo
<i>Melissodes robustior</i>		2	6	2.88	221	45.7	below	excavate	solitary	oligo
<i>Melissodes stearnsi</i>	4		69	2.04	216	66.6	below	excavate	solitary	oligo
<i>Melissodes tepida</i>	1	27		2.21	200	84.0	below	excavate	solitary	poly
<i>Nomada</i> sp. 1	2	3	5	2.20	UK	NA	below	NA	parasite	NA
<i>Nomada</i> sp. 2		1	1	1.37	133	NA	below	NA	parasite	NA
<i>Nomada</i> sp. 3	7	1	1	2.10	UK	NA	below	NA	parasite	NA
<i>Nomada</i> sp. 4		4		1.67	UK	NA	below	NA	parasite	NA
<i>Nomada</i> sp. 5		1	1	0.97	UK	NA	below	NA	parasite	NA
<i>Nomada</i> sp. 7			1	1.80	UK	NA	below	NA	parasite	NA
<i>Osmia aglaia</i>			2	2.20	139	NA	above	rent	solitary	poly
<i>Osmia atrocyanea</i>			6	2.82	131	NA	above	rent	solitary	poly
<i>Osmia bridwelli</i>			4	3.22	136.5	NA	UK	rent	solitary	UK
<i>Osmia cara</i>			1	3.07	133	NA	UK	rent	solitary	UK
<i>Osmia coloradensis</i>		1	5	2.31	136	NA	above	rent	solitary	oligo
<i>Osmia cyanella</i>			7	2.45	129	NA	above	rent	solitary	UK
<i>Osmia glauca</i>			2	1.61	119	NA	above	rent	solitary	oligo
<i>Osmia granulosa</i>		1	9	2.06	142	43.8	UK	rent	solitary	UK
<i>Osmia laeta</i>		2	3	1.95	136	NA	above	rent	solitary	poly
<i>Osmia montana</i>			2	2.49	137	NA	above	rent	solitary	oligo
<i>Osmia nemoris</i>	4	6	39	2.21	128.5	55.5	mix	rent	solitary	poly
<i>Osmia nigrifrons</i>			1	2.49	UK	NA	above	rent	solitary	oligo
<i>Osmia regulina</i>		1	12	1.98	143	NA	above	rent	solitary	poly
<i>Osmia</i> sp. "FA"			5	1.71	133	NA	UK	rent	solitary	UK
<i>Osmia</i> sp. "FC"			5	2.20	UK	NA	UK	rent	solitary	UK
<i>Panurginus nigrellus</i>		34	11	1.32	95	NA	below	excavate	solitary	poly
<i>Peponapis pruinosa</i>		14		3.13	184	NA	below	excavate	solitary	oligo
<i>Perdita heliotropii</i>			3	0.87	206.5	NA	below	excavate	solitary	oligo
<i>Protosmia rubifloris</i>		1	15	1.47	113	58.6	above	rent	solitary	poly
<i>Sphecodes</i> sp.	3	1		1.42	143	NA	below	NA	parasite	NA

<i>Stelis montana</i>		1	2.29	134.5	NA	above	NA	parasite	NA
<i>Triepeolus</i> sp. 1	1		UK	UK	NA	below	NA	parasite	NA
<i>Triepeolus timberlakeii</i>		1	1.71	148	NA	below	NA	parasite	NA
<i>Xeromelecta californica</i>	1		2.15	182.5	150.6	below	NA	parasite	NA
<i>Xylocopa tabaniformis</i>	1		5.84	128.5	99.0	above	excavate	solitary	poly